The Projective-Orbit Retry Theorem, Expanded

A detailed proof of Theorem 7.1 and a complete calculation of the unit-square distance matrix

Quantum Hodology Research Project

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Abstract

Theorem 7.1 of *Exact Low-Dimensional Hodology* gives a simple but powerful recipe: encode a group orbit in a quantum system, make a displacement of metric length r succeed with probability $1/r$, and charge one unit for every attempt. Retrying then costs r on average, and the triangle inequality proves that no adaptive sequence of other actions is cheaper. This note explains every assumption and every inequality in that argument. It also corrects a left-versus-right invariance convention that is harmless for the abelian examples in the paper but matters for nonabelian groups. Finally, it derives every entry of the four-by-four matrix D in Eq. (43), checks the Bellman equations directly, and performs the complete Schoenberg calculation showing that D is exactly the Euclidean metric of a unit square. Finally, it removes the assumption that the agent is handed its group label: a tetrahedral SIC outcome prepares and reports the current operational anchor, and SIC-outcome goals retain the same square geometry after subtraction of a derived common reporting overhead.

Contents

1 What the theorem is trying to accomplish

Suppose an agent has operational situations indexed by elements x of a group G. An action indexed by $h \in G$ attempts to translate the situation. The desired difficulty of displacement h is a prescribed group metric length

$$r_h = d_G(\mathsf{e}, h).$$

The construction turns this length into a waiting time. One attempt costs one intervention and succeeds with probability

$$p_h = \frac{1}{r_h}.$$

If failure changes nothing, repeated attempts have geometric mean waiting time $1/p_h = r_h$. This immediately supplies a policy whose expected cost equals the desired distance. The nontrivial half of the theorem is to prove that a clever combination of shorter displacements, stochastic choices, or history-dependent actions cannot do better.

The proof uses the metric distance to the goal as a *potential*:

$$V_g(x) = d_G(x, g).$$

The triangle inequality bounds how much any successful action can reduce this potential. The probability $p_h = 1/r_h$ is calibrated so that the *expected* decrease of V_g in one unit-cost intervention is at most one. Reducing an initial potential $V_g(x)$ to zero therefore costs at least $V_g(x)$ on average. Direct retry attains this bound.

2 The assumptions, one by one

2.1 A group of displacements

A group G has an identity e, inverses, and an associative product. The product records composition of displacements. For example, in the square construction

$$G = \mathbb{Z}_2 \times \mathbb{Z}_2,$$

with componentwise addition modulo two. Its elements are $00, 10, 01, 11$. Every nonidentity element is its own inverse.

2.2 A translation-invariant metric

A metric d_G is *left invariant* if

$$d_G(ax, ay) = d_G(x, y) \qquad \text{for every } a, x, y \in G,$$

and *right invariant* if

$$d_G(xa, ya) = d_G(x, y).$$

It is *bi-invariant* if it has both properties. On an abelian group the left and right translations coincide, so either invariance implies the other.

The normalization

$$\min_{h \neq \mathsf{e}} d_G(\mathsf{e}, h) = 1$$

is not cosmetic. It guarantees $r_h \geq 1$, hence $0 < p_h = 1/r_h \leq 1$, as required for a probability. It also fixes the unit of cost: the shortest nontrivial displacement costs one expected intervention.

2.3 A projective quantum orbit

A projective unitary representation assigns a unitary U_g to every $g \in G$, with multiplication respected up to global phase:

$$U_g U_h = e^{i\phi(g,h)} U_{gh}.$$

Global phase disappears under conjugation, so the density operators

$$\rho_g = U_g \rho_0 U_g^\dagger$$

carry an exact group action. The distinct-orbit assumption is

$$\rho_g = \rho_h \quad \Longrightarrow \quad g = h.$$

It says that two different group labels do not denote exactly the same physical density operator. It does *not* say that the states are orthogonal or perfectly distinguishable in one shot.

The cost proof itself is a statement about the induced controlled group process. Assumption (2.3) supplies its physical quantum realization and prevents an exact stabilizer from silently identifying two purported places.

2.4 The retry instrument

For every nonidentity displacement h, set

$$K_{\mathrm{s}}^{(h)} = \sqrt{p_h}\, U_h, \qquad K_{\mathrm{f}}^{(h)} = \sqrt{1 - p_h}\, I, \qquad p_h = \frac{1}{d_G(\mathsf{e}, h)}. \tag{1}$$

The labels s and f mean success and failure. This is a valid two-outcome quantum instrument because

$$\begin{aligned} K_{\mathrm{s}}^{(h)\dagger} K_{\mathrm{s}}^{(h)} + K_{\mathrm{f}}^{(h)\dagger} K_{\mathrm{f}}^{(h)} &= p_h U_h^\dagger U_h + (1 - p_h) I & (2) \\ &= p_h I + (1 - p_h) I = I. & (3) \end{aligned}$$

Conditional on failure, normalization removes the scalar and the state is unchanged. Conditional on success, normalization removes $\sqrt{p_h}$ and the state is conjugated by U_h.

3 A convention issue in the printed statement

The printed theorem lets success act by left multiplication, $x \mapsto hx$, but assumes only that d_G is left invariant. It then uses

$$d_G(x, hx) = d_G(\mathsf{e}, h).$$

For a general nonabelian group, that equality follows from *right* invariance, not left invariance: right multiplication of both arguments by x^{-1} sends (x, hx) to (e, h).

There are two equally valid repairs.

1. Keep the printed dynamics $x \mapsto hx$ and assume that d_G is right invariant. The direct displacement is $h = gx^{-1}$.
2. Keep a left-invariant metric but use right-action dynamics $x \mapsto xh$. The direct displacement is $h = x^{-1}g$.

The paper's exact examples use abelian groups such as $\mathbb{Z}_2^2$, $\mathbb{Z}^2$, and finite phase tori. Their metrics are bi-invariant, so the displayed proof and every reported result are unchanged. The distinction matters only when generalizing the theorem to nonabelian action groups.

For definiteness, the next section gives the corrected left-action/right- invariant form. Reversing every product gives the left-invariant/right-action form verbatim.

4 The theorem with a fully explicit proof

Theorem 4.1 (Projective-orbit retry construction, convention-correct form)**.** *Let G be finite or countable and let d_G be a right-invariant metric normalized by (2.2). Suppose G has a projective unitary representation and a fiducial density operator satisfying (2.3). Under action $h \neq \mathsf{e}$, failure leaves x fixed and success sends x to hx, with the instrument (1). Every attempt costs one. If the goal is g, then the optimal expected hitting cost is*

$$D(x, g) = d_G(x, g).$$

Proof. We separate the proof into validity, achievability, and optimality.

1. The quantum actions are valid. Equation (3) proves trace preservation after summing over the two reported branches. Because $d_G(\mathsf{e}, h) \geq 1$, both coefficients are real and lie between zero and one.

2. Direct retry achieves the metric distance. Assume $x \neq g$ and choose

$$h_* = gx^{-1}.$$

On success, $h_* x = g$, so one successful event reaches the goal. On failure, the state and group coordinate remain at x, so the same experiment starts again from exactly the same situation. The number T of attempts until the first success therefore has

$$\Pr(T = t) = (1 - p_{h_*})^{t-1} p_{h_*}, \qquad t = 1, 2, \ldots.$$

This is the geometric distribution, whose mean is

$$\mathbb{E}T = \frac{1}{p_{h_*}} = d_G(\mathsf{e}, h_*).$$

Right invariance, applied by multiplying both arguments by x, gives

$$d_G(\mathsf{e}, gx^{-1}) = d_G(x, g).$$

Thus direct retry has expected cost $d_G(x, g)$, proving

$$D(x, g) \le d_G(x, g). \tag{4}$$

3. No single action can beat the distance potential. Define

$$V(y) = d_G(y, g).$$

If the current point is y and action h is selected, the expected one-step cost plus future potential is

$$Q_h(y) = 1 + (1 - p_h)V(y) + p_h V(hy). \tag{5}$$

The reverse triangle inequality follows from the ordinary triangle inequality:

$$V(y) = d_G(y, g) \le d_G(y, hy) + d_G(hy, g) = d_G(y, hy) + V(hy).$$

Consequently

$$V(y) - V(hy) \le d_G(y, hy).$$

Right invariance gives

$$d_G(y, hy) = d_G(\mathsf{e}, h) = \frac{1}{p_h},$$

and hence

$$p_h\big[V(y) - V(hy)\big] \le 1. \tag{6}$$

Rearranging (5),

$$Q_h(y) = V(y) + 1 - p_h\big[V(y) - V(hy)\big] \ge V(y). \tag{7}$$

Thus every allowed action has Bellman cost at least the metric potential.

4. Why this also rules out adaptive policies and mixtures. Conditioned on the complete past, a policy may choose any action, possibly at random. Inequality (6) says that the conditional expected decrease of V during that intervention is at most one for every pure choice. Averaging preserves the inequality for a randomized choice. After n interventions, the expected total decrease is therefore at most n. For a policy with hitting time τ, apply this statement to $\tau \wedge n$ and let $n \to \infty$. At the goal, $V = 0$, so every proper policy of finite expected duration obeys

$$V(x) \le \mathbb{E}\tau.$$

Policies with infinite expected duration cannot improve the optimum. Hence

$$D(x, g) \ge V(x) = d_G(x, g). \tag{8}$$

Combining (4) and (8) proves equality. $\square$

4.1 Where equality occurs

For the direct action $h_* = gx^{-1}$, success lands at g, so $V(h_*x) = 0$, while $1/p_{h_*} = V(x)$. Therefore

$$Q_{h_*}(x) = 1 + \left(1 - \frac{1}{V(x)}\right) V(x) = V(x).$$

All alternative actions satisfy $Q_h(x) \geq V(x)$; some may tie, but none can be smaller. This is the precise sense in which the triangle inequality is the optimality certificate.

5 Eq. (43): the unit-square example

5.1 The group and its desired metric

Take

$$G = \mathbb{Z}_2 \times \mathbb{Z}_2 = \{00, 10, 01, 11\}.$$

Addition is componentwise modulo two. We associate these labels with the four Euclidean points

$$00 \leftrightarrow (0,0), \quad 10 \leftrightarrow (1,0), \quad 01 \leftrightarrow (0,1), \quad 11 \leftrightarrow (1,1). \tag{9}$$

Define the translation-invariant group metric by its displacement lengths:

$$d_G(\mathsf{e}, 10) = 1, \qquad d_G(\mathsf{e}, 01) = 1, \qquad d_G(\mathsf{e}, 11) = \sqrt{2}, \tag{10}$$

and

$$d_G(x, g) = d_G(\mathsf{e}, g - x).$$

Here subtraction equals addition because every element has order two. The only nontrivial triangle inequality is $\sqrt{2} \leq 1 + 1$; hence this is indeed a metric. Since G is abelian, it is both left and right invariant.

5.2 The qubit orbit

The physical system is a qubit with fiducial Bloch vector

$$\boldsymbol{r}_{00} = \frac{1}{\sqrt{3}}(1, 1, 1).$$

Conjugation by Pauli X and Z generates four states:

$$\begin{aligned} \boldsymbol{r}_{00} &= (+,+,+)/\sqrt{3}, &\qquad \boldsymbol{r}_{10} &= (+,-,-)/\sqrt{3}, \\ \boldsymbol{r}_{01} &= (-,-,+)/\sqrt{3}, &\qquad \boldsymbol{r}_{11} &= (-,+,-)/\sqrt{3}. \end{aligned}$$

They are distinct and form a tetrahedron on the Bloch sphere. Every distinct pair has overlap

$$\mathrm{Tr}(\rho_x \rho_g) = \frac{1}{3}.$$

Thus their *physical-state geometry* is equidistant. The square below is instead the geometry of optimal controlled cost.

5.3 Removing the hidden-location assumption with the SIC instrument

The labels $00, 10, 01, 11$ should not be supplied as an unobserved location register. The tetrahedral orbit itself defines a four-outcome qubit SIC. Let $\Pi_o = \rho_o$ be its rank-one projectors and define

$$M_o = \frac{1}{\sqrt{2}}\Pi_o, \qquad o \in \{00, 10, 01, 11\}. \tag{11}$$

Because a qubit SIC obeys $\sum_o \Pi_o = 2I$,

$$\sum_o M_o^\dagger M_o = \frac{1}{2}\sum_o \Pi_o = I.$$

Thus $\{M_o\}$ is a valid four-outcome instrument.

If its input is orbit state ρ_x, then

$$p(o \mid x) = \mathrm{Tr}(M_o \rho_x M_o^\dagger) = \frac{1}{2}\,\mathrm{Tr}(\Pi_o \Pi_x) = \begin{cases} 1/2, & o = x, \\ 1/6, & o \neq x. \end{cases} \tag{12}$$

The outcome does not reveal the *premeasurement* orbit label perfectly; four nonorthogonal qubit states cannot be perfectly discriminated. The key fact is instead the conditional update

$$\frac{M_o \rho_x M_o^\dagger}{p(o \mid x)} = \Pi_o. \tag{13}$$

Every possible outcome o prepares the corresponding tetrahedral state. Immediately after the SIC action, the agent therefore knows its current operational state: it is the last observed outcome token. There is no hidden coordinate inference in this statement. The binary strings are convenient theorist names and may be replaced by arbitrary colors or integers.

The agent can learn what opaque movement buttons do as well. Starting from an observed SIC anchor o, it presses a button and performs the same SIC again. The next-outcome law has its $1/2$ peak at the translated anchor and $1/6$ elsewhere. Repeating this common test recovers the induced permutations of the four outcome tokens, up to the unavoidable choice of origin, axes, and signs. Thus both "where I am" and "what this action does" can be grounded in observed statistics rather than privileged binary coordinates.

5.4 The three displacement actions

The axial displacements 10 and 01 have length one, so their retry probabilities are one. They are deterministic Pauli channels:

$$10 : \rho \mapsto X\rho X, \qquad 01 : \rho \mapsto Z\rho Z.$$

The diagonal displacement 11 is implemented by Y, projectively equivalent to XZ, and has

$$p_{11} = \frac{1}{\sqrt{2}}.$$

Therefore

$$K_{\mathrm{s}} = \sqrt{p_{11}}Y = 2^{-1/4}Y, \qquad K_{\mathrm{f}} = \sqrt{1 - \frac{1}{\sqrt{2}}}I. \tag{14}$$

Repeated diagonal attempts have expected cost

$$\frac{1}{p_{11}} = \sqrt{2}.$$

Taking two deterministic axial steps would cost 2, so direct diagonal retry is better. The theorem proves that no adaptive alternative costs less than $\sqrt{2}$.

5.5 Computing every entry of D

Order rows and columns as $00, 10, 01, 11$. The diagonal entries vanish because the agent is already at its goal:

$$D(00,00) = D(10,10) = D(01,01) = D(11,11) = 0.$$

Pairs differing in exactly one bit are axial neighbors and cost one:

$$\begin{aligned} D(00,10) &= 1, & D(00,01) &= 1, \\ D(10,11) &= 1, & D(01,11) &= 1. \end{aligned}$$

Pairs differing in both bits require displacement 11 and cost $\sqrt{2}$:

$$D(00,11) = \sqrt{2}, \qquad D(10,01) = \sqrt{2}.$$

The metric is symmetric, so reversing any pair gives the same entry. Hence

$$\boxed{D = \begin{pmatrix} 0 & 1 & 1 & \sqrt{2} \\ 1 & 0 & \sqrt{2} & 1 \\ 1 & \sqrt{2} & 0 & 1 \\ \sqrt{2} & 1 & 1 & 0 \end{pmatrix}.} \tag{15}$$

The same calculation can be displayed as a displacement table:

source	goal	group difference	optimal expected cost
00	10	10	1
00	01	01	1
00	11	11	$\sqrt{2}$
10	01	11	$\sqrt{2}$
10	11	01	1
01	11	10	1

6 A direct Bellman check for the square

It is useful to verify one column of D without invoking the theorem as a black box. Choose goal $g = 11$. From (15), the proposed values are

$$V(00) = \sqrt{2}, \qquad V(10) = 1, \qquad V(01) = 1, \qquad V(11) = 0.$$

At 00, deterministic X or Z gives

$$Q_X(00) = 1 + V(10) = 2, \qquad Q_Z(00) = 1 + V(01) = 2.$$

The diagonal action succeeds at 11 with probability $1/\sqrt{2}$ and otherwise remains at 00, so

$$\begin{aligned} Q_Y(00) &= 1 + \left(1 - \frac{1}{\sqrt{2}}\right) V(00) + \frac{1}{\sqrt{2}} V(11) \\ &= 1 + \left(1 - \frac{1}{\sqrt{2}}\right) \sqrt{2} \\ &= 1 + \sqrt{2} - 1 = \sqrt{2}. \end{aligned}$$

Thus the diagonal action attains the proposed value and the axial alternatives are more expensive.

At 10, the deterministic Z action reaches 11 in one step, so $V(10) = 1$. Similarly, X sends 01 to 11 in one step. At the goal, the process terminates with value zero. The other goal columns follow by translation symmetry. This explicitly verifies the stochastic-shortest- path Bellman equations whose solution is (15).

7 The stronger operational goal: obtain a SIC outcome

We can now eliminate even the assumption that a goal means occupying a hidden group label. Let goal g mean: perform the SIC action and observe outcome token g. Every movement and every SIC report costs one intervention. If a SIC outcome is not g, the episode continues from the newly prepared state Π_o. After the first SIC outcome this is a fully observed finite MDP, because the most recent token is a sufficient operational state.

By square symmetry, write v_0, v_1, v_2 for optimal cost when the current anchor is respectively the goal, an axial neighbor, or the opposite corner. The proposed policy reports at the goal, moves deterministically from an edge to the goal, and retries the diagonal displacement from the opposite corner. The two movement rules give

$$v_1 = 1 + v_0, \qquad v_2 = \sqrt{2} + v_0. \tag{16}$$

At the goal state, the SIC reports g with probability 1/2, terminating the episode. Its two edge outcomes have total probability 1/3, and its diagonal outcome has probability 1/6. Consequently

$$v_0 = 1 + \frac{1}{3}v_1 + \frac{1}{6}v_2 \tag{17}$$

$$= 1 + \frac{1}{3}(v_0 + 1) + \frac{1}{6}(v_0 + \sqrt{2}), \tag{18}$$

so

$$\boxed{v_0 = c := \frac{8 + \sqrt{2}}{3}}. \tag{19}$$

It follows that the full reported-goal value matrix is

$$\boxed{V_g(x) = c + D(x, g).} \tag{20}$$

The remaining Bellman inequalities confirm optimality. At the goal, moving first costs $c + 2 > c$. At an edge, reporting immediately costs

$$1 + \frac{2}{3}v_1 + \frac{1}{6}v_2 > v_1,$$

while the axial move attains v_1. At the diagonal, reporting costs

$$1 + \frac{1}{3}v_1 + \frac{1}{2}v_2 > v_2,$$

and an axial two-step route costs $c + 2 > c + \sqrt{2}$; diagonal retry attains v_2. Thus (20) is the optimal Bellman solution, not merely the value of a selected policy.

The nonzero self-value $V_g(g) = c$ is the common overhead of requesting a stochastic SIC report. Hodological separation is obtained by subtracting this goal-specific self baseline:

$$V_g(x) - V_g(g) = D(x, g). \tag{21}$$

The exact Euclidean square therefore survives when places and goals are both defined entirely by experienced SIC outcomes.

Before the first SIC action, an agent with an unknown preparation occupies an epistemic initial condition rather than one of four known anchors. Its first SIC outcome both reports a token and prepares the associated state. Subsequent tracking uses only that token, chosen actions, and observed success/failure events; the policy needs no concealed binary-coordinate variable.

8 Exact computation and finite-sample simulation

The companion program `sic_anchored_square.py` constructs all density matrices and Kraus operators, solves the four stochastic-shortest-path problems, runs Monte Carlo navigation, and learns randomly renamed Pauli permutations from anchor–action–SIC experiments. Five focused tests verify Kraus completeness, branch preparation, the $1/2, 1/6$ response kernel, the analytic Bellman solution, and optimal-policy structure.

At seed 20260813, exact value iteration gives

$$\max_{x,g} |V_g(x) - c - D(x,g)| = 2.58 \times 10^{-14},$$

and baseline subtraction recovers D to 9.10×10^{-15}. Across 50,000 Monte Carlo episodes for each of the 16 ordered pairs, maximum mean-cost error is 0.02831. In 500 independent opaque-label trials per sample size, the complete four-action response permutations are recovered in every trial at 100 samples per anchor/action pair.

9 Why D is exactly a two-dimensional Euclidean metric

The entries of D already match the familiar side and diagonal lengths of a unit square. Schoenberg's test certifies this without presupposing the coordinates in (9).

First square every entry:

$$D^{\circ 2} = \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 \\ 1 & 2 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{pmatrix}. \tag{22}$$

Every row mean, every column mean, and the grand mean of this matrix equal one. With

$$J = I - \frac{1}{4}\mathbf{1}\mathbf{1}^T,$$

the double-centered Gram matrix is

$$B = -\frac{1}{2} J D^{\circ 2} J \tag{23}$$

$$= \begin{pmatrix} \frac{1}{2} & 0 & 0 & -\frac{1}{2} \\ 0 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & 0 & 0 & \frac{1}{2} \end{pmatrix}. \tag{24}$$

To see the geometry directly, center the square coordinates:

$$X = \begin{pmatrix} -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}. \tag{25}$$

Multiplication gives $XX^T = B$. Therefore B is positive semidefinite. The two columns of X are orthonormal, so the eigenvalues of B are

$$1,\ 1,\ 0,\ 0.$$

It has rank two, and classical multidimensional scaling reconstructs the four rows of X, up to rotation and reflection. This proves intrinsically that the hodological metric (15) is an exact two-dimensional Euclidean metric.

10 What the theorem proves—and what it assumes

The theorem proves three useful facts.

1. The Kraus operators form valid quantum instruments.
2. The desired metric is achieved by a concrete policy.
3. The triangle inequality is a global certificate against every adaptive shortcut.

It does not derive the metric from an otherwise neutral quantum dynamics. The desired length is inserted into the success law $p_h = 1/d_G(\mathsf{e}, h)$. In the square, the Euclidean diagonal is obtained because the diagonal action was assigned success probability $1/\sqrt{2}$. A different probability would produce a different hodological diagonal.

It also does not imply that instrument outcomes localize the agent. In a random-unitary retry action, success/failure reports whether a chosen translation occurred; it need not reveal the starting state. Finally, the four qubit states form a tetrahedron in Bloch-state geometry even though their optimal costs form a square. This separation is intentional: hodological distance measures difficulty of controlled goal achievement, not quantum state overlap.

The SIC extension in Sections 5.3 and 7 repairs the location-token problem in a precise sense: an outcome prepares and therefore identifies the *postmeasurement* anchor, and place goals are future SIC outcomes. It does not retrospectively identify the unknown premeasurement state, nor make the retry movement outcome itself location-informative. Its virtue is that the agent's current-state token and its goal tokens now arise from one actual measurement alphabet.

11 A compact checklist for using the theorem

Before applying the retry theorem to a new construction, check:

1. Is every desired displacement represented by a valid group element?
2. Is the orbit faithful after quotienting global phase and the fiducial stabilizer?
3. Does the side of multiplication match the side of metric invariance?
4. Has the metric been normalized so every $p_h \le 1$?
5. Does failure truly leave the operational coordinate unchanged?
6. Is the goal defined by the physical orbit, a predictive equivalence class, or an external group-product monitor? These are not interchangeable.
7. Is $p_h = 1/d_G(\mathsf{e}, h)$ physically motivated, learned, or simply supplied by design?

For Eq. (43), all algebraic conditions hold exactly. The remaining scientific limitation is the last one: the Euclidean success law is engineered rather than derived.

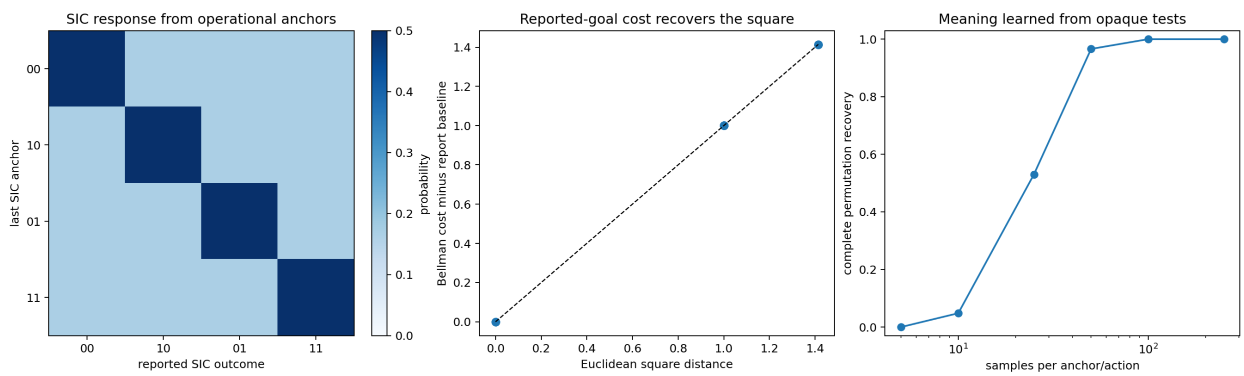

Figure 1: **The square grounded in SIC outcomes.** Left: after an observed SIC anchor, the same outcome has probability 1/2, every other outcome has probability 1/6, and every branch prepares the state named by its token. Center: subtracting the common reporting baseline from exact Bellman costs recovers side 1 and diagonal $\sqrt{2}$. Right: randomly renamed Pauli response permutations are learned from future SIC tests without exposing binary coordinates.